Chapter 4
AI in Child Welfare and Family Services

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1 Introduction

Child abuse and neglect, or child maltreatment, is a pressing and costly public health issue with negative lifelong consequences. According to the nationwide administrative data (U.S. Department of Health & Human Services, 2025), during the federal fiscal year 2023, about 7.8 million children received child protective services investigations for child abuse and neglect in the United States. Of these, about 546,159 cases were confirmed, and 2000 children died from child abuse and neglect. While all child maltreatment is detrimental and should be prevented and addressed, child neglect is the most pervasive form of maltreatment. Of the 546,159 confirmed cases of maltreatment, approximately 75% were due to neglect. This means that out of the 546,159 confirmed cases, around 409,000 were instances of child neglect. It is important to parse out the difference between categories of maltreatment, given the elusive nature of child neglect and the challenges associated with the use of risk assessments to predict the same. The field has historically come under scrutiny for its inability to differentiate child neglect from family poverty (Gibbs et al., 2024; Milner & Kelly, 2020) and the resulting harm that such failures cause families. As will be discussed later in this chapter, AI and assessment technology are often used not only to identify those children most at risk for maltreatment but also to make critical judgments about resource and service allocations. Unlike other forms of child maltreatment, child neglect is often intertwined with a range of socioeconomic conditions, many of which child welfare and family services (CWFS) have limited capacity to change in any substantive way. Given the

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complexities inherent in child neglect—many of them socio-structural in nature—yet the predominantly individual-level focus of most interventions, we must consider how to use AI in ways that fully capture the challenging dynamics of prediction in this context.

Child welfare and family services (hereafter, CWFS) provide a continuum of services to ensure that children are safe and that families have the support they need to care for them successfully. These services typically include receiving and investigating possible child maltreatment reports, providing services to families that need assistance and support in the safety and care of their children, arranging for children to live with foster families or relatives when they are considered to be not safe at home, and arranging for reunification, adoption, or other permanent family placements for children and youth leaving foster care. During each step of the process, CWFS workers are expected to document their observations and decisions. At some points, they also need to retrieve historical information to help determine which service options to choose.

While helpful in most respects, the CWFS system has historically been criticized for administrative burdens, reporting inconsistencies, and disparities in service and resource distribution. Racial disparities in families' involvement in the system, such as the overrepresentation of Black/African American and Native American children in the system, have been at the center of the debate (Dettlaff & Boyd, 2020; Edwards, 2025; Drake et al., 2023). While these critiques and discussions are not new, researchers and policymakers have been calling for efforts and reforms that can identify risk earlier, ensure supports are provided with precision to those in need, and ultimately prevent system involvement and child maltreatment for families at risk. Recognizing the potential to make unbiased, evidence-based decisions, CWFS is increasingly integrating artificial intelligence (AI) tools into its decision-making and service-delivery systems to improve risk assessments and reduce human bias. AI-assisted tools, such as machine learning technologies, help CWFS workers identify potential child maltreatment cases and support them in making informed decisions. While many current AI applications have shown some success, they also face significant challenges and criticisms.

To critically examine the current applications, benefits, challenges, and ethical considerations of AI in CWFS, we conducted a comprehensive literature review on AI tools used in CWFS. We recognize that there are existing literature reviews on AI and CWFS (Gibbs et al., 2024b; Hall et al., 2024). However, these studies generally focus on a specific area overlapping with AI use, such as predictive and prescriptive analytics in child welfare or the birth match policy. Also, we would like to acknowledge the use of AI tools in the copyediting process of this chapter. However, this chapter is fundamentally human-centered, developed through the efforts of the authors and research assistant students who conducted the literature search, review, and writing. Below, we review the use of AI in CWFS by answering the following questions: How is AI currently utilized in child welfare and family services? What are the benefits associated with its implementation? What challenges and concerns arise from AI's use in this field? What are the future directions and implications of AI use in child welfare and family services?

2 AI in Child Welfare and Family Services: Current Applications

Over the past decades, CWFS has increasingly used AI across various operations, including risk assessment, child abuse and neglect detection, and service recommendations such as placement decisions and matching children with foster parents. Below, we detail the current uses of AI in CWFS, highlighting applications, AI models, data sources, and operational settings.

2.1 Risk Assessment and Decision Support

In the context of CWFS, risk assessment typically refers to predictive risk modeling that evaluates the likelihood that a child may experience maltreatment in the future (Hall et al., 2024). This process considers factors related to the children, their family, and their broader environment. When a relatively higher risk is identified, CWFS agencies can respond by prioritizing support and resources for families most in need, ideally intervening before harm occurs. Modern predictive risk modeling is a growing approach that uses machine learning methods to analyze large, diverse datasets, such as social service records, health care information, and demographic data, to estimate the probability of adverse outcomes related to child maltreatment (Schwartz et al., 2017). Unlike traditional statistical models, which are often built around pre-defined variables and assumptions, AI techniques such as machine learning models are more flexible. They can learn and adapt over time, helping identify new patterns and relationships as more data becomes available. These AI-driven models can incorporate numerous variables and are often used for so-called prescriptive analytics (Hall et al., 2024), which goes beyond predicting outcomes to guide decision-making by suggesting potential actions. For this section, we focus specifically on studies that implemented sophisticated, data-driven machine learning algorithms in CWFS. While we acknowledge that the field also uses more traditional tools, such as actuarial checklists and structured decision-making instruments, we did not include those in this review. These conventional tools are typically based on a fixed set of historical and current risk factors and do not evolve with new data inputs (Gleeson, 1987).

Currently, AI-assisted risk assessment tools are being increasingly used in the CWFS sector across the United States and internationally. A 2021 survey by the American Civil Liberties Union found that 26 U.S. states have considered using predictive analytics in child welfare, while jurisdictions in at least 11 states are actively doing so (Samant et al., 2021). A well-known example that has received extensive attention in research and practice is Allegheny County, Pennsylvania, which implemented a predictive risk model to assess hotline reports (Allegheny County Department of Human Services, 2019). The Allegheny Family Screening Tool is an AI-assisted tool introduced to Allegheny County, PA's Office of Children, Youth, and Families Intake/Call Screening Department. It has since been adapted in many other states. The tool was designed to assist child protection hotline call screeners in assessing child maltreatment risk by providing a risk score indicating the likelihood that a child may be placed out of the home in 2 years. The AI model processes historical administrative data and generates risk scores based on predictive analytics. The model analyzes data from various sources, including previous child maltreatment reports, family demographics (e.g., socioeconomic status, family history), criminal records of parents or guardians, substance abuse and mental health history, and emergency hotline call records. Another example is California's predictive analytics model in Los Angeles County, which was modeled closely on the Allegheny Family Screening Tool but used only historical administrative data that the child welfare agency already held to improve screening decisions for child protective service investigations (Putnam-Hornstein et al., 2022). Other examples in the U.S. CWFS agencies include Florida (Schwartz et al., 2017).

Common features of the above tools include adaptive algorithms trained to predict future maltreatment risks using an extensive database, including prior case histories and social service reports. These assessment tools demonstrate how AI-based risk assessment tools can improve risk prediction and support CWFS caseworkers. However, they also generated concerns about how to avoid biases, ensure fairness, and maintain human oversight. For example, in 2017, Illinois's Department of Children and Family Services discontinued the use of a predictive analytics tool due to its ineffectiveness in accurately identifying children at high risk of harm (Drake et al., 2020). In the sections below, we will detail concerns about false positives arising from AI-assisted tools.

2.2 AI-Assisted Service Recommendations

AI also supports CWFS providers in service planning and decision-making, such as out-of-home placement decisions and matching children with foster parents, by analyzing data patterns and predicting outcomes. Out-of-home placement decisions are particularly important for a child's well-being, as placement instability and frequent moves can lead to adverse outcomes, including delayed permanency, poorer educational outcomes, and increased behavioral and mental health issues. During the federal fiscal year 2022, a total of 569,879 children were served by the foster care system (U.S. Department of Health and Human Services, 2023). With this large number, CWFS has been exploring ways to improve placement outcomes. There are many factors to consider in CWFS decision-making (Chor, 2013), including child safety, family dynamics, and service availability, making AI techniques an ideal approach to addressing this challenge by leveraging multiple data sources. Currently, the most used tools by states to inform placement recommendations include the Child and Adolescent Functional Assessment Scales (Hodges & Kim, 2000), Child and Adolescent Level-of-Care Utilization System (Sowers et al., 2003), Child and Adolescent Needs and Strengths (Lyons, 2008), and Treatment Outcome Package

(TOP; Kraus et al., 2005). The first three tools are mostly traditional, rule-based, human-scored assessment tools that use scoring rubrics to inform placement recommendations. The TOP, however, received some modern implementations, including machine learning for early detection of poor emotional and behavioral outcomes. TOP provides a multidimensional assessment of children's behavioral health and well-being across multiple domains from various informers. A recent study used TOP's detailed assessments as critical input features to develop machine learning models that can predict the probability of treatment success in psychiatric residential care versus other placements (Trudeau et al., 2023). By integrating TOP into their methodology, the study demonstrates the promise of machine learning for predicting positive placement outcomes and informing future placement decisions for youth in care. Overall, while promising, studies are still rare in using AI-assisted tools for out-of-home placement recommendations.

Child-foster parent matching focuses on the compatibility between a child's specific needs and a caregiver's capabilities. Existing research consistently indicates that matching children with foster parents who are emotionally involved and well-prepared to meet children's behavioral and emotional needs can lead to increased stability and well-being for children (Carnochan et al., 2013). By analyzing past placement data, caregiver profiles, and child characteristics, AI can recommend stable matches. Recent research shows that a web-based matching algorithm implemented in Kansas helps pull together child and placement information to inform placement decisions. The study shows the promise of the matching system in improving permanency and reducing disruptions by using temperament and caregiving styles (Moore et al., 2016). However, the specific computational methods employed by the algorithm are not detailed in the available literature. Although there are potential benefits, AI-assisted child-foster parent matching remains under-researched.

2.3 Detection of Child Abuse and Neglect

Although AI implementation in child abuse and neglect detection remains limited compared to other fields, its applications are expanding. Image recognition models, particularly deep learning techniques, have been used to distinguish abuse from non-abuse in children based on the radiology reports of head computed tomography, brain MRIs, skeletal surveys, and child self-figure drawing (Tsai & Kleinman, 2022). Other tools, such as natural language processing, have also been used to distinguish child abuse from non-abuse based on electronic medical records (Annapragada et al., 2021). For example, a recent study (Shahi et al., 2021) explored the application of deep learning and natural language processing to identify cases of child physical abuse. The researchers used clinical notes and radiology reports from pediatric hospital data to train deep learning models. The models showed that AI could assist clinicians in identifying potential cases of abuse more efficiently and accurately. There are also studies focused on text mining and machine learning to detect child maltreatment or domestic violence from unstructured data, such as narrative case reports (Amrit et al., 2017; Victor et al., 2021). Collectively, these studies underscore the growing role of AI in detecting child maltreatment. However, despite the positive results, the implementation of such tools is still rare.

2.4 AI-Assisted Training for CWFS Practice

AI has also been used to train CWFS professionals, especially in investigative interviewing training. Trainees in CWFS often require opportunities to practice in realistic settings with immediate feedback (Lamb, 2016). Importantly, AI-driven avatars enable the simulation of child interviewees, offering dynamic conversational practice and AI-generated feedback. Several recent studies have explored the creation of AI-powered avatars as realistic, conversational CWFS training tools designed to simulate interactions with children who may have experienced maltreatment (Baugerud et al., 2021; Salehi et al., 2022; Hassan et al., 2022a, 2022b). These technologies provide professionals a safe, controlled, and consistent environment to build interviewing skills, receive feedback, and improve child-centered communication. For example, Salehi et al. (2022) explored how synthesized avatars can be used to develop questioning skills in trainees. The study demonstrated that these avatars could simulate realistic dialogue and provide feedback in line with best interview practices. Hassan et al. (2022a) further integrated a virtual reality environment with a talking avatar. Together, these tools have contributed to a growing body of research on AI-powered training tools that can address the shortage of practical training opportunities in investigative interviewing.

3 Benefits of AI in Child Welfare and Family Services

When properly trained, AI technologies have shown promise in assisting decision-making, reducing operational costs, and delivering better outcomes for children and families. Studies have applied AI-assisted predictive analytics to examine child maltreatment-related outcomes, such as foster care placement, maltreatment re-reports, aging out of out-of-home care, birth outcomes, service use of child-welfare-involved caregivers, and cases requiring judicial or preventive intervention (Ahn et al., 2021; Chor et al., 2025; Chouldechova et al., 2018; Janczewski & Nitkowski, 2023; Schwartz et al., 2017; Vaithianathan et al., 2020). These studies all showed promising results in the predictive power of AI-assisted tools. Below, we summarize several key benefits of AI in CWFS from the existing literature.

3.1 *Data-Driven Decision-Making with Reduced Human Bias*

A key advantage of AI in CWFS is its predictive power that can provide early identification of families and children in need of services and supports before harms escalate. Traditional human decision-making in CWFS has been criticized for the influence of implicit biases on outcomes (Miller et al., 2013). Although racial disproportionality in child welfare system involvement is widely documented, the factors contributing to these issues have been the subject of debate over the past decades (Dettlaff & Boyd, 2020; Drake et al., 2023; Edwards, 2025). Clearly, however, the debate regarding these factors has led to efforts to address human biases that may particularly affect families of color and those from lower socioeconomic backgrounds. When trained on diverse, representative datasets, AI models can reduce subjectivity by offering data-driven, evidence-based recommendations. Data-driven AI techniques also uncover complex, multidimensional patterns in data that traditional statistical methods may miss. Research assessing AI tools in risk prediction has shown that machine learning techniques could significantly improve the accuracy and utility of the risk assessment instrument (Hall et al., 2024). For example, a recent study in Broward County found that AI tools significantly improved accuracy in identifying cases likely to re-enter the system. The study found that 57% of recurrences occurred within 1 year of case closure, and AI models were effective at flagging these high-risk cases (Schwartz et al., 2017). These tools also assisted in classifying cases requiring judicial action, preventive services, or intensive intervention. By improving risk differentiation, AI-enabled systems helped match services more effectively to family needs, potentially improving child safety and long-term well-being.

3.2 *Improved Case Management and Administrative Efficiency*

CWFS workers often need to make decisions and provide service within a tight timeline with potentially life-changing consequences for children and families. Also, the administrative burden on CWFS workers is overwhelming, as they must manage large volumes of case files and sometimes perform repetitive tasks. Without immediate access to critical information and with the emotional strain, errors can arise that carry serious consequences. As a result, families may be wrongly targeted, children at risk may be overlooked, and workers may face heightened burnout, all of which can strain public trust and resources. At worst, failures in decision-making can jeopardize child safety.

AI tools have the potential to address these challenges by reducing administrative workloads, automating data entry and analysis, reducing manual errors, and expediting case assessments. Predictive risk models, as mentioned above, analyze historical case data, case worker assessments, and demographic information to estimate the likelihood of child abuse or neglect (Vaithianathan, 2019). These tools can instantly analyze large volumes of data, from the point of hotline intake to service recommendations, using system-stored information available before a call is made. Unlike humans, AI tools can scan all relevant records without fatigue or oversight, which ensures no data point is overlooked. These tools enable CWFS workers to access critical data promptly and allocate resources proactively, thereby reducing false negatives and false positives. AI tools also bring efficiency benefits. By automating repetitive tasks, AI reduces administrative burden and allows social workers to focus more on family engagement than paperwork (Gillingham, 2020). These tools streamline workflows, enhance data access, and support real-time decision-making, thereby improving both worker efficiency and service delivery in emotionally charged, information-rich environments.

3.3 Strategic Service Recommendation and Resource Allocation

In addition to minimizing biases and human errors in decision-making, AI models can effectively allocate CWFS resources when making service recommendations, such as out-of-home placement decisions and child-foster parent matching. Machine learning algorithms offer decision support to CWFS workers by providing data-driven recommendations. These tools analyze an extensive database of case histories and provide insights into the best service or intervention based on past successful cases. By offering case recommendations based on previously successful interventions, these AI-based tools also improve consistency and reduce variability in assessments and, eventually, achieve equitable outcomes (Trudeau et al., 2023). AI also helps direct CWFS workers, financial support, and prevention programs to where they are most needed by predicting which geographic regions or populations are at higher risk. For example, deep learning models can forecast surges in neglect or maltreatment cases. These systems, integrated with Geographic Information Systems, can locate high-risk communities using crime statistics, economic indicators, and historical child welfare data. Still, as stated in other sections of this chapter, the use of the technology in this manner should be done judiciously, given the biased assumptions often made about children and families whose neighborhoods or communities of origin, demographics, or other macro conditions depict a picture that may not accurately convey their unique experiences and needs. Theoretically, by predicting which communities may have a high risk of child maltreatment, resources such as shelters, foster care placements, and counseling services can be allocated more effectively. While this is an aim of the technology, we know that help-seeking and the receipt of needed services can be frustrating and elusive for many families.

3.4 *Facilitating Communication and Training Tools*

AI-powered chatbots and virtual assistants can also provide immediate support to both families and CWFS professionals. These systems provide access to parenting resources, legal information, and crisis support, helping families navigate services in real time. Regarding CWFS professionals, AI tools, such as chatbot-based training tools, help them practice critical skills in child interviews in an interactive environment. These interactive tools support emotional, cognitive, and behavioral engagement among professionals and enable continuous learning and skill retention (Røed et al., 2023). In addition, although primarily studied outside CWFS, developing engaging AI-supported platforms can help educators prepare students for careers in high-pressure, real-world settings.

4 Concerns and Challenges

While studies generally embrace the strengths of AI in CWFS, its implementation also raises a host of concerns across technical and ethical domains (Gillingham, 2020; Saxena & Guha, 2024). Many of these concerns are not only hypothetical but have been demonstrated in real-world cases, as evidenced by critiques from researchers, practitioners, and communities that have experienced unintended harms and challenges arising from the use of algorithmic systems in CWFS services. Complicating the full embrace of AI in child welfare is a level of practice tension that predates technology and is not necessarily caused by AI. Failure to recognize this tension can stagnate progress or increase skepticism about AI in child welfare. The three current points of tension are crucial for understanding how AI is perceived and approached.

Practice Dilemma 1: The existence of a remove-and-place system and the absence of a support system. AI advancement in child welfare must be examined within the contested goals of Child Welfare and Family Assistance (CWFA) systems. The child welfare system aims to protect, promote well-being, and ensure permanency (U.S. Department of Health and Human Services, Administration for Children and Families, 2023). However, critics argue that the system overemphasizes family surveillance and child removal, focusing less on providing services to families in need. This belief suggests the system punishes parents by removing children, even when challenges stem from socio-structural phenomena beyond individual control. Some scholars differentiate abuse and neglect and, in doing so, advocate for a different response system for child neglect (Feely et al., 2020). Other advocates call for policy-level interventions to provide economic security and reduce maltreatment rates (Berger & Slack, 2021). Additionally, care coordination and cross-system collaboration have long been advanced as mechanisms to improve child and family outcomes (Mariscal et al., 2024). AI advancements that coordinate services across the continuum of care are needed. AI can be valuable when data and learning models account for broader environmental conditions and identify appropriate response patterns. Predicting adverse conditions without providing practitioners with tools to respond and support families limits progress.

Practice Dilemma 2: Inadequacy of child removal. AI as an assessment aid raises concerns about its impact on human decision-making. Predictive analytics may lead child welfare professionals to substantiate maltreatment and remove children, but this may not improve decision-making. Proponents argue that AI tools offer objectivity, aiding difficult decisions, while skeptics question whether this is, in fact, the very problem. About 50% of children removed by child protective services are reunited within a short period (USDHHS-ACF, 2023). This issue is problematic due to the family disruption, trauma, and inherent bias and institutionalized racism it produces (Merritt et al., 2025; Roberts, 2014, 2019, 2022). High-touch case management and authentic engagement with children and parents should be service-delivery objectives. Reconciling AI use in critical service decisions requires demonstrating its myriad purposes, downplaying its current use as a marker for child removal.

Practice Dilemma 3: The need to predict and respond to adverse community conditions. Families in the child welfare system are often deemed inadequate rather than suffering from strained community conditions. Decision-makers should prioritize child and family well-being. Decision-making tools frequently fail to account for the community adversity associated with foster care risks (White-Wolfe & Denby, in press). Increased reliance on AI tools in child welfare settings will occur when they decipher pathways to reverse economic hardships and structural issues associated with child welfare involvement. AI could help explain racial disparities in child welfare and formulate policies to reduce harm to impoverished or minoritized families.

Below, we summarize several additional key concerns and challenges raised in the literature.

4.1 Data-Driven Bias and Unintended Consequences

One of the most significant challenges in applying AI to CWFS is the potential for algorithmic bias. AI systems are objective only when the data on which they are trained is objective. If the data already embed existing biases, the AI system will accurately predict them and ultimately reproduce systematic inequalities. For example, many risk-prediction tools are highly dependent on administrative data (Putnam-Hornstein et al., 2022). When the historical child welfare data reflects deep-rooted systemic racial/ethnic and socioeconomic inequities, predictive risk models may inadvertently reinforce these disparities rather than correct them. Although scholars have proposed a range of fairness metrics, such as calibration, predictive parity, and error rate balance, these definitions are often incompatible with one another in practice (Chouldechova et al., 2018; Corbett-Davies & Goel, 2018; Dwork et al., 2012). A tool might be equally accurate across two racial groups but still have a higher false-positive rate for one group. In the CWFS context, this can translate into disproportionate scrutiny or intervention for certain populations, such as Black and African American families.

Despite growing awareness, relatively few child welfare studies explicitly assess bias and fairness during model development (Chouldechova et al., 2018; Lanier et al., 2019; Ahn et al., 2021; Purdy & Glass, 2023; Chor et al., 2025). A recent review study found that less than half of the reviewed published work in the child welfare field considered ethics, equity, or participatory design principles (Hall et al., 2024). Also, administrative data often contain errors, inconsistencies, or duplicate records (Vogl, 2020; Allard et al., 2018). Without robust mechanisms for uniquely identifying individuals across systems, data can be fragmented or misleading, further undermining the reliability of AI predictions. While AI may help deduplicate records, current systems struggle when individuals withhold information or when only partial data is available. These challenges are especially problematic at the frontline level, where decisions made with flawed data can have life-altering consequences for children and families.

In fact, while many child welfare agencies continue to explore the AI tools in CWFS, several state or local agencies, including Illinois, Los Angeles County, and, most recently, Oregon, have shut down predictive analytics programs due to concerns over poor outcomes and racial bias. In Allegheny County, as we mentioned above, while the AI-assisted family screening tool supported objective decision-making, it was also found to produce biased risk scores, which caseworkers frequently had to override (Kawakami et al., 2022b). Ironically, the tool meant to reduce bias may end up adding layers of complexity and mistrust.

4.2 AI Misalignment with Human Judgement

Although AI tools are designed to assist practitioners' decision-making, studies indicate a persistent misalignment between the long-term predictive goals of AI systems and the immediate, context-driven needs of CWFS professionals. For example, in a 2-year ethnographic study with child welfare agencies, researchers found that algorithms predicting long-term risks may not provide actionable insights for addressing immediate family needs, leading to skepticism and reduced trust in these tools (Saxena & Guha, 2024). Many studies found that caseworkers often report that AI tools do not reflect their professional judgment (Kawakami et al., 2022a, 2022b; Saxena et al., 2020). In some cases, algorithmic design choices, such as aggregating household-level risk or including static variables like criminal justice involvement, have amplified systemic discrimination rather than mitigated it. CWFS practitioners expressed frustration when predictions are based on proxy indicators, such as prior child welfare system involvement, that feel misaligned with their day-to-day decision-making (Saxena & Guha, 2024). Also, tools such as the Child and Adolescent Needs and Strengths assessment require workers to collect and enter detailed data, often without providing them with increased decision-making power in return. As a result, AI was initially intended to reduce workload and improve efficiency, but in many cases, it has added new layers of data entry and administrative burden for caseworkers (Saxena & Guha, 2024).

4.3 Regulatory and Theoretical Gaps

Despite the growing adoption of AI tools, the CWFS field lacks clear regulatory frameworks to guide the ethical development and use of these tools (Saxena & Guha, 2024). There are few federal or state-level guidelines on how such tools should be used, evaluated, or overseen. This lack of guidelines creates significant ethical risks, particularly in a field already marked by racial and socioeconomic disparities. Without a clear regulatory framework, CWFS and AI tool developers also must navigate the process in a gray zone, which can raise concerns about data privacy and bias that reinforce systemic inequalities. Equally important is the absence of a firm theoretical grounding. A review study found that, among 50 studies on computational systems used in CWFS, only one used a theoretical model grounded in the child welfare literature (Saxena et al., 2020). The review study found that several studies have acknowledged this gap and proposed frameworks, such as cumulative risk models, causal models, or revised SDM models grounded in risk and resilience theory, to conduct research in AI and child welfare. Yet, such models have not been implemented within existing literature. Without theoretical frameworks that meaningfully guide the practice of AI tools, AI tools may risk oversimplifying the multidimensional factors influencing family well-being into binary classifications based on risk scores. For example, the authors argued that many AI tools used predictors based on readily available, quantifiable child and family characteristics. Yet, many such factors, such as the severity of abuse, have little indication that they are related to the recurrence of abuse (Saxena et al., 2020). Also, such factors are often based on individual- or family-level characteristics rather than on broader social contexts, such as structural or historical factors. The absence of interdisciplinary, child welfare-specific theoretical models only reduces the validity and fairness of AI tools and limits their potential to support preventive approaches in CWFS. In short, without models that reflect the realities of family life and systemic challenges, AI tools risk oversimplifying complex social issues.

4.4 Human Trust and Transparency

Beyond technical issues, AI introduces ethical concerns around consent, surveillance, and stigmatization (Drake et al., 2020). Many families are unaware that their data is being used in algorithmic systems, raising important questions about privacy and confidentiality. When data is collected for one purpose and used for another without informed consent, it not only undermines public trust but also risks harming those it was intended to help. This inevitably led to a trust issue, which remains a significant barrier to successful AI integration in CWFS. For technology to support, not replace, human decision-making, both users and the public must fully understand how the system works, how to interpret its outputs, and when to question or override its recommendations. While the public provides a mixed response to the use of AI in CWFS, the trust within CWFS itself has been mixed. Research has found that practitioners are generally skeptical and distrustful of AI's judgment (Kawakami et al., 2022a). When workers do not trust these tools, organizational pressures may push them to rely on AI recommendations to avoid liability or to comply with policy requirements. The significant impact of the algorithm-generated score on their decision-making has been shown not only among CWFS professionals but also in the legal context (Trail, 2024). Therefore, while external transparency (e.g., public documentation of a tool's design) is essential, internal transparency and training are equally important. Also, the lack of transparency in how these tools operate has confused CWFS practitioners (Kawakami et al., 2022a). Many practitioners reported not receiving adequate training or information on how the models function and how the system responds to different inputs (Gibbs et al., 2024a). Taken together, workers need ongoing support to engage meaningfully with AI tools, and agencies need to foster environments that respect professional judgment alongside technological guidance.

5 Recommendations and Future Directions

Overall, despite the concerns around the use of AI tools in CWFS, most studies suggested these concerns can catalyze thoughtful, interdisciplinary dialogue and innovation, not as a reason to abandon the potential of these technologies altogether (Drake et al., 2020; Hall et al., 2024; Lanier et al., 2019). If designed and implemented with care, transparency, and collaboration, AI can help strengthen decision-making, reduce disparities, and ultimately improve outcomes for children and families. Below, we provide a summary of key recommendations to guide future work based on the current literature review.

5.1 Improve Accuracy and Account for Human Misalignment

While predictive tools should be accurate, it is equally important to recognize that accuracy alone is not enough. In the predictive risk model context, for instance, studies have suggested that differences in prediction targets led to misalignment between AI recommendations and workers' actual decision-making objectives (Drake et al., 2020; Kawakami et al., 2022a). Comparing human decisions to algorithmic outputs without understanding the underlying purpose of each can be misleading. If an algorithm misidentifies families due to structural biases (e.g., poverty, racial disparities), it risks reinforcing the very inequities the child welfare system seeks to address (Drake et al., 2020). Some studies have highlighted the importance of human oversight in algorithmic decision-making, such as in reducing racial disparities (Cheng et al., 2022). While algorithms, such as those in the predictive risk model, can provide valuable data-driven insights, incorporating human judgment is also crucial for interpreting these recommendations within appropriate social and historical contexts. Furthermore, any AI system used in CWFS needs to undergo comprehensive and ongoing evaluation (Chor et al., 2025; Lanier et al., 2019). It is essential to ask whether predictive tools are reducing rates of serious harm, such as fatal maltreatment, in the jurisdictions where they have been implemented. These evaluations should be based on high-quality longitudinal data and incorporate causal inference methods to determine whether the tools are meeting their intended goals or unintentionally causing harm.

5.2 Build Internal and External Transparency

Transparency is also needed to be both intentional and continuous. This means engaging families, communities, and CWFS practitioners throughout the design and use of AI tools and clearly documenting the rationale, development, and implementation processes. Allegheny County serves as a promising example of external transparency, with publicly available documentation detailing how its predictive model was built, validated, and monitored (Allegheny County Department of Human Services, 2019). However, internal transparency remains an issue. Workers need to understand the tools they are expected to use, such as what data drives them, how scores are generated, and when to trust (or question) their outputs. True transparency also means creating space for public oversight and community dialogue. Jurisdictions should solicit feedback from affected families, frontline workers, and advocacy groups before and after implementation. Those families' lived experiences must also inform technologies that impact families. It is critical to involve parents, caregivers, and children, especially those from historically marginalized communities, in conversations around AI-assisted tools such as predictive screening and service recommendations.

Notably, there is an apparent disconnect between the number of AI systems built and how frontline professionals work. Many studies suggest that most AI-assisted tools lack a grounding in child welfare theory and are rarely co-designed with the people who use them daily (Kawakami et al., 2022a). They have recommended bridging AI and human-computer interaction through participatory design (Ahn et al., 2024b; Gibbs et al., 2024; Hall et al., 2024). This approach actively involves CWFS workers in both the design and evaluation of tools, thereby ensuring that systems are aligned with their needs and professional judgment. Professionals expected to use these tools, including social workers and healthcare providers, should also be part of shaping how systems are designed, monitored, and refined. Engaging these groups ensures a deeper understanding of intended benefits and unintended consequences and supports more ethical, inclusive policymaking. Many child welfare agencies lack in-house data science capacity. One way to close this gap is to invest in interdisciplinary partnerships among CWFS organizations, computer science departments, and policy schools (Hall et al., 2024; Kum et al., 2015). These collaborations could support training, research, and even the creation of internship pipelines for students in CS, informatics, and social work. Extending Title IV-E funding to support workforce development in AI and data-informed practice could also equip CWFS professionals with the skills needed to engage critically with these tools. For example, understanding model performance and its implications, such as how 75% accuracy might sound impressive in a research paper but fall short in high-stakes decisions, requires both technical and contextual expertise.

5.3 Promote Equal and Ethical Frameworks

Finally, research suggested that we need to ensure that systems work equitably and ethically across all groups (Ahn et al., 2024a; Drake et al., 2020). Any tool that produces significantly higher false-positive rates for a specific racial/ethnic group or socioeconomic group is ethically problematic. Concerns about equity are not new in child welfare. They arise at every decision point, from screening and investigation to service delivery and out-of-home care placement. While AI may help reduce child maltreatment or improve risk assessment, it also introduces new risks, such as over-surveillance or biased intervention (Saxena & Guha, 2024). These trade-offs should be made explicit and debated publicly. Predictive tools should be audited and evaluated regularly to assess performance across subpopulations, and adjustments should be made to address disparities when they appear. Additionally, algorithmic design needs to consider legal and ethical boundaries, such as limits on the use of sensitive information (e.g., juvenile justice records and histories of abuse) in prediction models. Regarding data privacy and consent, some scholars argued that AI tools, such as predictive risk models, should be treated similarly to other methods. Therefore, in cases where the child welfare agency has granted data sources use, the use of those data sources would not pose new ethical concerns (Drake et al., 2020). While debates continue over this issue, policies related to risk assessment should also address accountability, transparency, and fairness.

6 Conclusion

Taken together, AI can play an influential role in CWFS, but only if implemented with care, transparency, and a commitment to equity. It is not enough for these tools to be accurate. They should also be fair, ethical, and aligned with the needs of both families, children, and the professionals who serve them. Achieving this balance requires not only technical expertise but also meaningful collaboration across disciplines and communities. That said, it is essential to ensure that these AI systems do not replicate the systemic biases already embedded in historical data. To address this, researchers are exploring fairness-aware machine learning and ethical AI governance frameworks that can improve transparency and accountability in CWFS (e.g., Ahn et al., 2021; Purdy & Glass, 2023). The use of AI in CWFS should never be about replacing human judgment. It should be about supporting it. By thoughtfully integrating ethical design, rigorous evaluation, inclusive collaboration, and transparent governance, we can move toward AI systems that serve children and families, especially those who have historically been underserved or harmed by the system.

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